Control ACS 80



,

8 Control system setup ACS 80

8.1 Activating frequency converter control unit

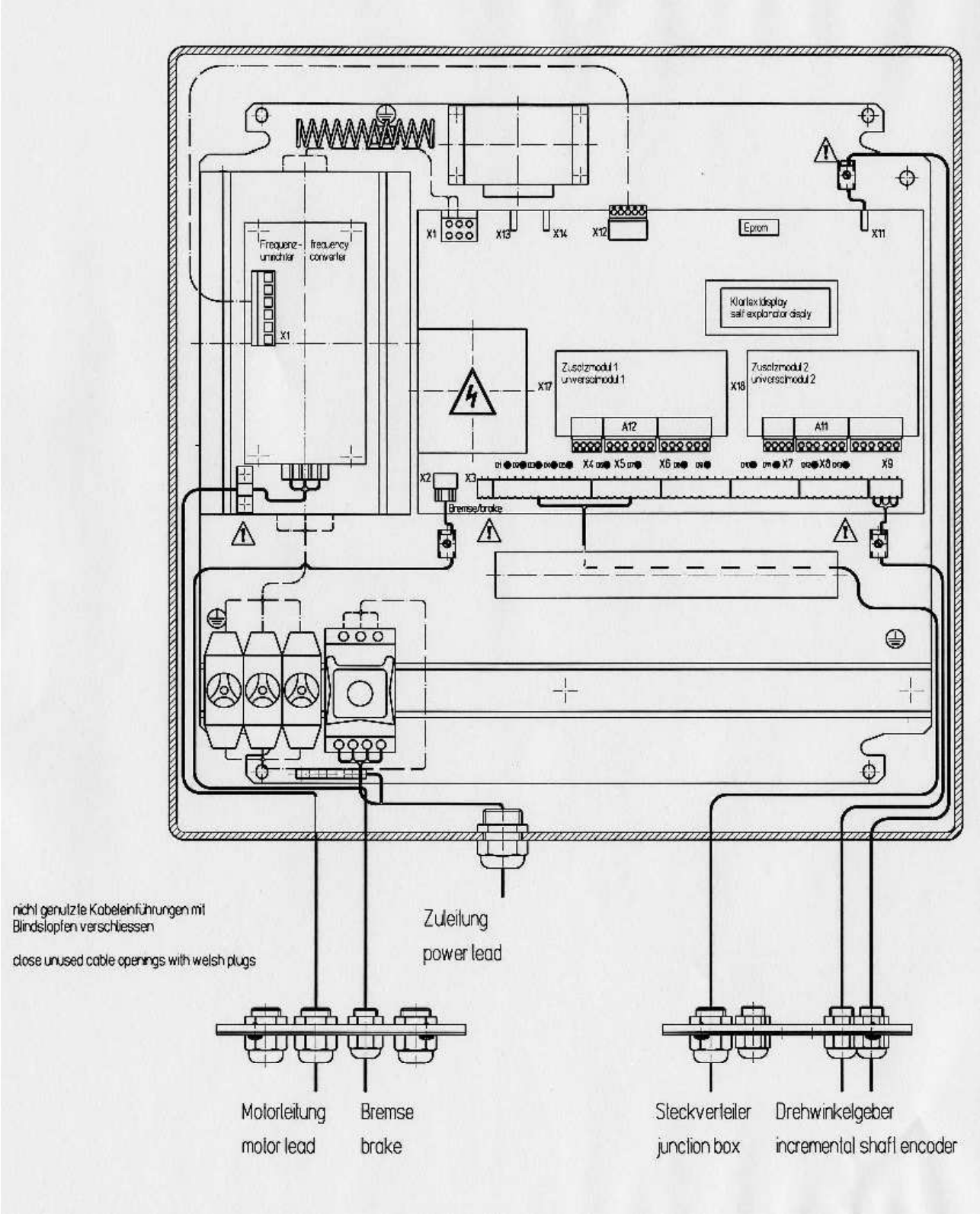
8.1.1 Preparations/Information

Before starting the initial operation of the FU control unit the following preparations must be made:

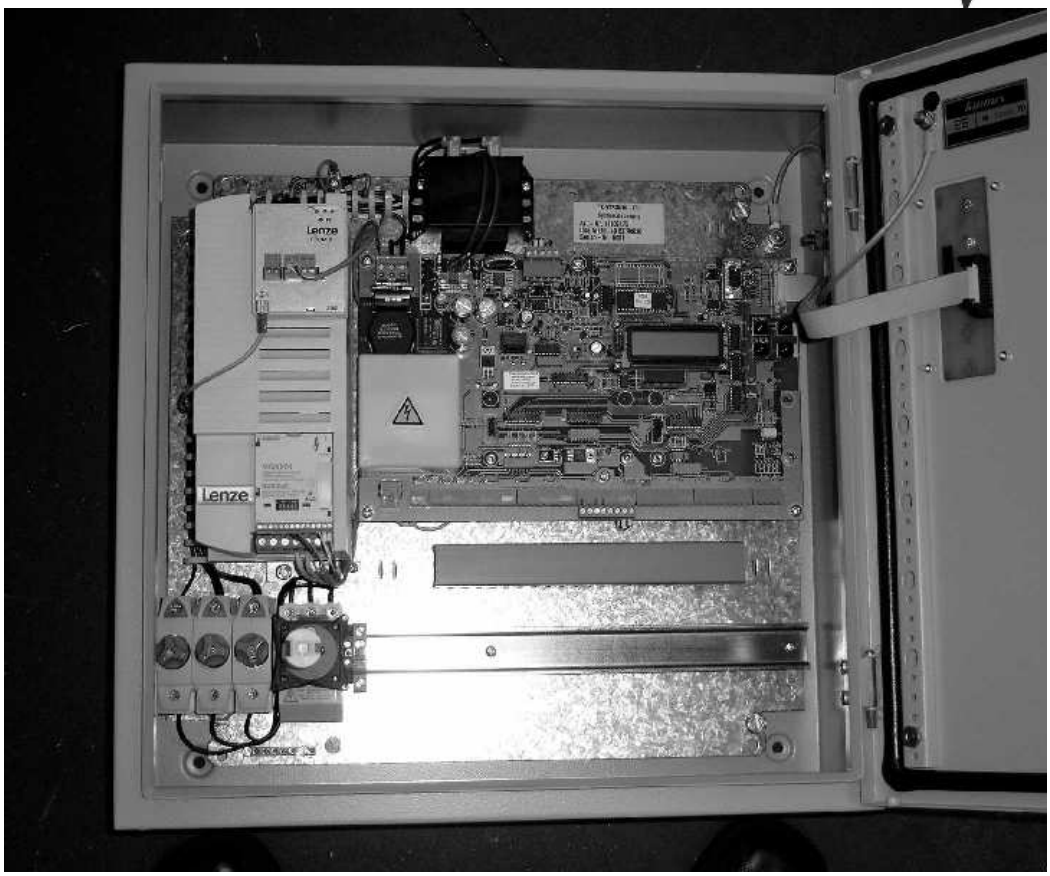
- Make sure that there will be no traffic of persons or cars within the vicinity of the door during initial operation.
- The door is installed according the installation manual.
- All connection power lines between door and control unit are installed.
- All external Impulse activators and safety features are connected.
- Unused inputs should be bridged according the wiring diagram.
- Power should be active. The door must be connected to the equipotential bonding.
- If a FI-safety switch is used it must be an all-current-sensitive type (Typ B). Only this type controls alternating currents and pulsing DC fault current. However, this safety switch should not be the only safety feature for indirect touch (VDE 0160). In case of doubt please get advice from a skilled electrician.
- The main switch is to be switched off.
- The door should be half open.

The connection cables should be installed according the following drawing. The cables should be put in a right angle. The power lines should be separately installed from the control cables, otherwise it could lead to malfunctions. Put plugs with corresponding protection category on unused inputs.

Connecting the shielded cables (motor, brake, incremental shaft encoder and airlock interconnection cables) please pay utmost attention on good large-surface contact of the shielding with the mounting base. Otherwise there may be errors in function of the door!



Principal design and cable installation of the ACS 80



Principal design of the ACS 80-2

To connect the shield the coating of the cable must be removed. Do not damage the shield! The now open metal shield is fixed with the screwed on clip at the mounting base or the enclosed shield clip. The unshielded cable between shield clip and control unit should be as short as possible. The shield clip for the motor cable is at the heatsink of the frequency converter (at ACS 80) or on the installation plate (ACS 80-2).

To avoid unauthorized changes of the control function a coded plug (dongle) is necessary for the initial installation. Switch off the main switch and plug in this plug into the corresponding piece X11. After switching on the main switch again the next steps will be shown by the LCD-window. The initial installation of the FU control unit is done via menu lead.

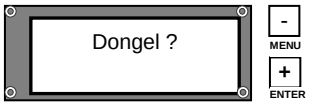
The selection of single function or parameters is done through the "open" (S3)- resp. the "close" (S4)- button at the terminal. With activating the "ENTER" (S2) button the pre-selected status is saved.

The button "MENU" (S1) switches to the next point of the menu. You may go through the menu by using this button without changing any saved values.

The control unit checks some of the set values automatically for it's plausibility! If the set values do not make any sense, for instance the upper end position is the same as the bottom end position, the initial operation mode can not be left. Also the removal of the programming plug does not lead to the finish of the initial operation mode. Only a repeated go through all points of the menu and saving of the correct values leads to a finish of the initial operation mode.

8.2 Initial operation

By using a new never programmed control system, you will see the following message after the first turn on in the display.

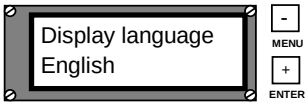


You will see this message only at the first programming. If you don't get it please start with 8.2.1.

Please turn of the power, plug on the dongle on plug X11 and turn on the power again.

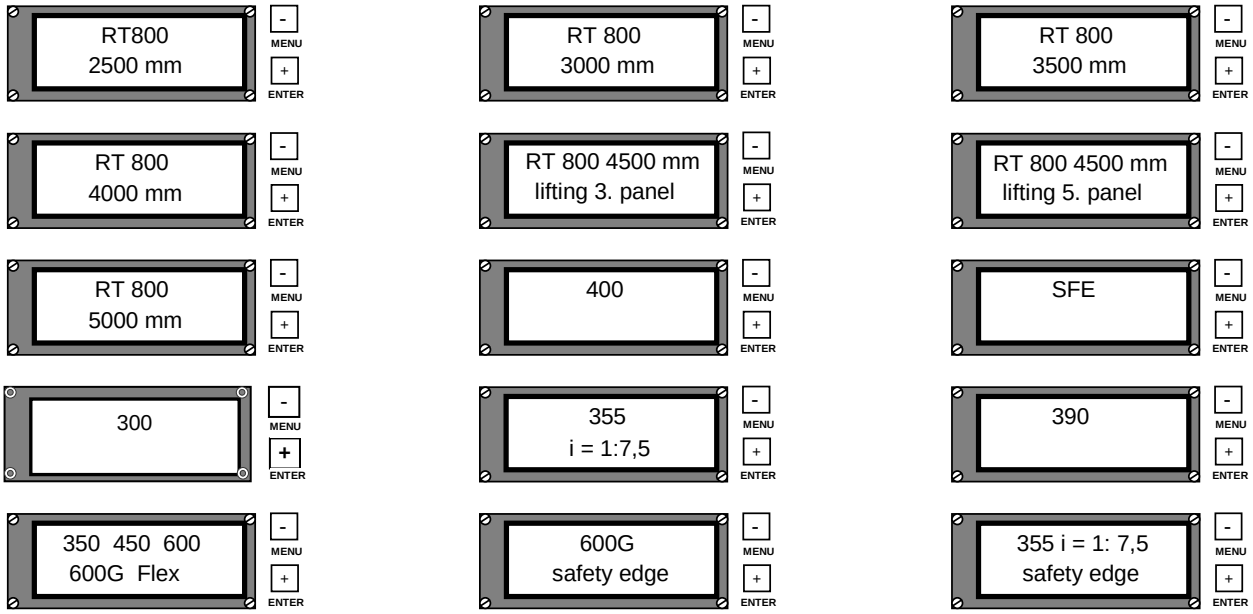
8.2.1 Display language

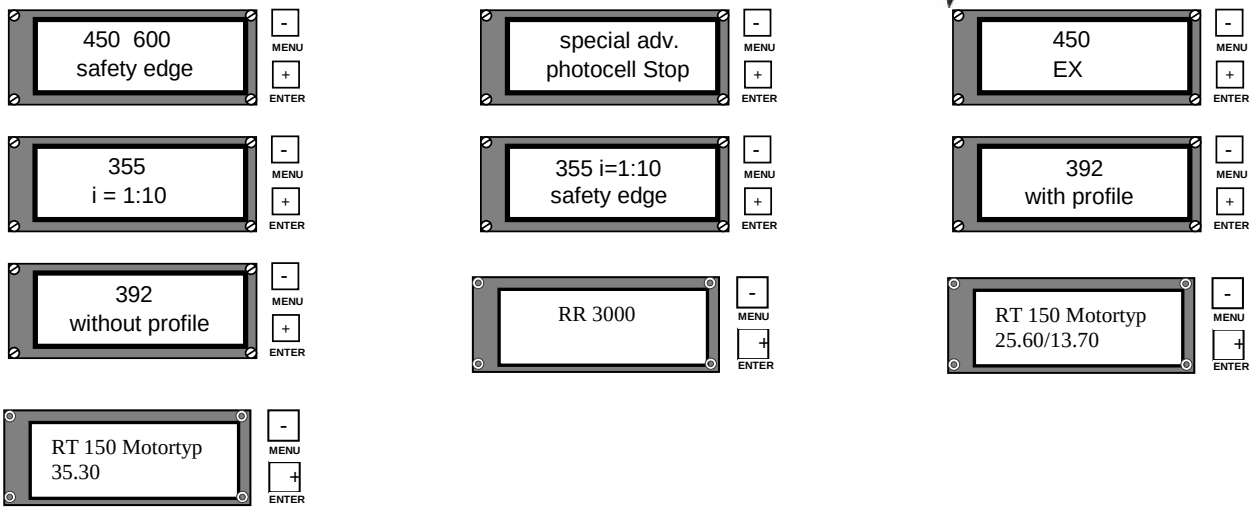
The main switch is turn on and after a few seconds the first point of the menu shows at the LCD-display.



The display language may be selected via the two arrow buttons. Possible are: German, English and Spanish. This menu point can be saved with the „ENTER" button. You may jump to the next point of the menu without saving the value with the „MENU" button. Herewith the former adjustment is being transferred.

8.2.2 Door type selection





Because of the used software version and the control variant (ACS 80 / ACS 80-2) different display messages may show up. For the adjustment of the door type pay attention please that for the RT 800-doors the correct height of the door is selected. For the 4500 mm high RT 800-door the driven segment (3. or 5.) must be additionally selected. Existing is an additional possibility of an in-between grid dimension (250 mm grid). In this case always select the next higher door type. For instance at a door height of 2750 mm the door type RT 800, 3000 mm must be selected. At the 355 door you must distinguish between the different gear multiplication's (i = 1:7,5 or 1:10). If the door type 392 is equipped with wind bars you must select the door type „392 with profile". Otherwise selection is made for „392 w/o profile". The doortype RT 1000 is the same as RT 3000.

If the door type „SFE", „300",„450 EX" or "RT 150" is selected, there will be no requests for the end positions. The end positions must then be set at the limit switch of the drive.

The menu point „door type selection" is being jumped over at the initial operation if the door has been preset with special parameters at manufacturers side.

The following selection only shows if the door type „400" has been selected (special option).

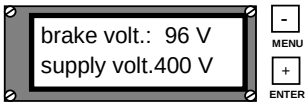


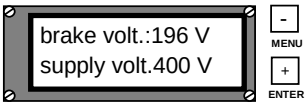
An operating current brake function as brake if voltage is on. A fail-safe-brake is lifting if on voltage.

8.2.3 Brake voltage

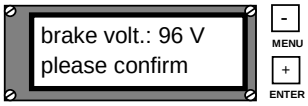
Here the connecting voltage of the brake must be selected. The value of the current power is stated at the last page of the wiring diagram. Furthermore you can read the voltage on the brake after dismantling the fan cover from the motor.

A wrongly set brake current may lead to destruction of the brake.





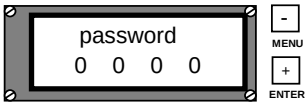
The selection may be done with the arrow buttons.
Following the selection the next point of the menu shows up:





This point of the menu must be confirmed with the „ENTER“ button. By activating the „MENU“ button the program jumps again into the selection of the brake current.

8.2.4 Special functions



This point of the menu is exceptionally for manufacturer pre-settings. With the “MENU” button you may switch to the next point.

8.2.5 FU-Parameterization

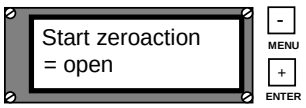


Here the necessary parameters are automatically transfered to the frequency converter. The transfer takes about 2-3 seconds and must not be confirmed. During transfer a red and a yellow LED are blinking (D46 und D47).

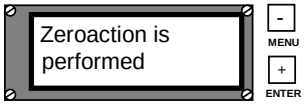
8.2.6 End position adjustment

During the initial operation the door can be manually run past the bottom end position (in hold-to-run-mode). This may lead to heavy damages on the door.

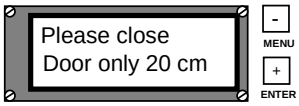
During the adjustment of the end position the door only runs in creeping speed.



If the door is correctly installed and is not located already in the upper end position it opens itself by activating the "OPEN" button to the reference limit switch. The following information shows

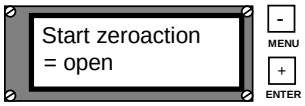


If the door is already complete open and the reference limit switch is occupied the following information shows



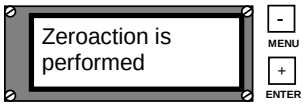
At this point of menu >if the door is run manually over the bottom end position it may lead to heavy mechanical damages of the door. The door may now be closed with the "CLOSE" button to a maximum of about 20 cm.

Now the following information shows

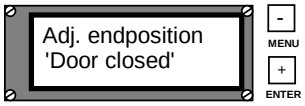


A following "Open" impulse opens the door up to the reference limit switch.

This limit switch gives the control unit the reference point of the door from which the impulse is being counted. At the same time it serves as safety limit switch. During zero action of the door (reference run) the following information shows



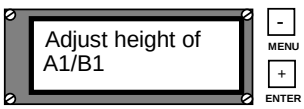
After reaching the reference limit switch the information deletes and the control unit automatically switches to the next point of the menu.

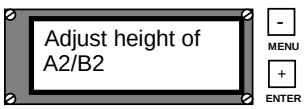


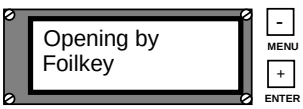
The lowest allowed bottom end position is reached if the middle part of the seal from the rubber bottom profile reaches the height of the base plate of the columns.

Because of the low speed during the creep run the bottom end position should be set for operation about 20 mm above the target position (set point). Since higher speed during regular operation results in a higher postrun distance and the bottom end position shifts the bottom end position slightly downwards. An exact adjustment depends on the door type and must be adapted at site.

With the buttons "OPEN" and "CLOSE" the door is to be taken to the end position "DOOR CLOSED". The setting is made in button-touch operation. This means; as long as a button is being touched the door moves into the corresponding direction. After letting go of the button the door stops immediately. Is the door near the required end position the fine adjustment can be made exactly to the mm by briefly touching the corresponding button. If the required end position is reached the adjustment must only be confirmed by activating the "ENTER" button. After that it automatically switches to the next point of the menu.







With the FU – control unit three different opening distances may be adjusted. Firstly follow the adjustments of the external impulse inputs A1/B1 and A2/B2.

The impulse activator inputs A1/A2 are directed to one door side and the impulse activator inputs B1/B2 to the other side. This way for example can A1 be connected to

the pull switch and A2 to an induction loop for different heights. (A and B for direction, 1 and 2 for different opening heights)

After that the opening height/distance is adjusted which is to be reached by activating the "OPEN" button on the key pad at the control box. The adjustment of the end positions "DOOR CLOSED" the required distances can be reached and safed with the buttons "OPEN" and "CLOSE". The maximum opening height/distance is given by the reference limit switch. Of course may the same distance be set for all impulse activator inputs.



A selected brake distance which is to short may lead to an overrun of the upper end position. A selected brake distance which is to long leads to an unnecessary timely prolongation of the opening movement and overheating of the motor.

This adjustment fixes where the brake ramp should be started before the upper end position. Here the number of impulses is set before reaching the corresponding end position. The value depends on the door type and the door size. As starting value you should select the following and then optimize the adjustment. Optimization of the value follows after the first initial operation. The preset values are only rough guidelines, an exact adjustment should be made during initial operation at the site. Adjustable are values between 50 and 400.

The brake distance is optimally set if the door reduces speed in even motion and then goes into end position by reaching the low speed. A too long brake distance leads to a long creep run.

	RT 800 2500mm	RT 800 3000mm	RT 800 3500mm	RT 800 4000mm	RT 800 4500 mm (lifting 3. panel)	RT 800 4500 mm (lifting 5. anel)	RT 800 5000mm	400	SFE
Brake distance up with ACS 80-control	300	240	300	300	220	300	300	300	Limit switch

	300	355 i = 1:7,5	390	350, 450, 600, 600G,Flex	600G safety edge	355 i=1:7,5 safety edge.	450,600 safety edge	special adv. photocell Stop
Brake distance up with ACS 80-control	Limit switch	140	160	250	250	140	250	240

	450 EX	355 i = 1:10	355 i = 1:10	392 w/ profile	392 w/o	RR 3000	RT 150 Motortype 35.30	RT 150 Motortype 25.60/

			Kontl.		profile			13.70
Brake distance up with ACS 80-control	Limit switch	230	230	180	250	300	Limit switch	Limit switch



A selected brake distance which is to short may lead to an overrun of the upper or bottom end position. A selected brake distance which is to long leads to an unnecessary timely prolongation of the closing movement and to overheating the motor.

This adjustment fixes the point where the brake ramp should be started before the bottom end position. Here the number of impulses is set before reaching the corresponding end position. The value depends on the door type and the door size. As starting value you should select the following and then optimize the adjustment. Optimization of the value follows after the first initial operation. The preset values are only rough guidelines, an exact adjustment should be made during initial operation at the site. Adjustable are values between 50 and 200.

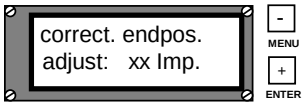
The brake distance is optimally set if the door reduces speed at even motion and then goes into end position by reaching the low speed. A too long brake distance leads to a long creep run.

	RT 800 2500mm	RT 800 3000mm	RT 800 3500mm	RT 800 4000mm	RT 800 4500 mm (lifting 3. panel)	RT 800 4500 mm (lifting 5. panel)	RT 800 5000mm	400	SFE
Brake distance bottom w/ ACS 80 control	20	20	20	20	20	20	20	50	Limit switch

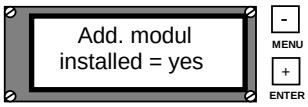
	300	355 i = 1:7,5	390	350, 450, 600, 600G,Flex	600G safety edge	355 i = 1:7,5 safety edge	450,600 safety edge	special adv. photocell Stop
Brake distance bottom w/ ACS 80-control	Limit switch	40	40	40	40	40	40	40

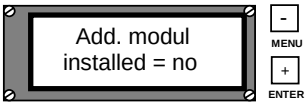
	450 EX	355	355	392	392	RR 3000	RT 150 Motortype	RT 150 Motortype

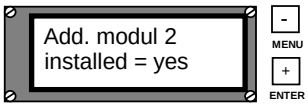
		i = 1:10	i = 1:10 Kontl.	w/ profile	w/o profile		35.30	25.60
Brake distance bottom w/ ACS 80-control	Limit switch	40	40	40	40	40	Limit switch	Limit switch

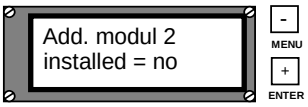


Here the fine adjustment of the bottom end position is possible. The adjustment value is between 0 and 30. Whereby the value from 0 to 7 leads to lower end positions and value from 9 to 30 leads to a higher end position. At adjustment value 8 the bottom end position remains as saved in the menu point „End position door adjustment“.
At the initial operation this value should for the time being be set to 8 so the true adjusted end position remains. Corrections can be made after the first initial operation with the help of this point in the menu. For this the point of menu „End position door closed“ does not need to be changed!





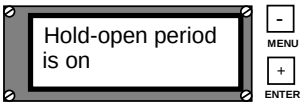


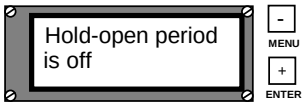


2 additional modules can be plugged onto the control terminal if required.
If these additional modules are installed they must be selected here.

8.3. Special functions

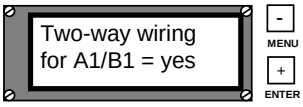
8.3.1 Keep open time

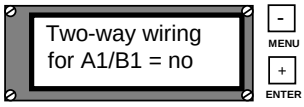


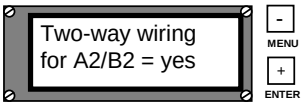


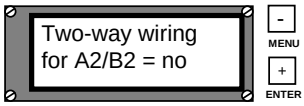
With this point of the menu the keep open time may be switched on or off. At switched on keep open time the door remains open according the set time and closes again automatically. This time may be changed between 2 and 99 seconds during regular operation hours. If automatic closing is not desired the keep open time must be switched off.

8.3.2 Toggle circuit





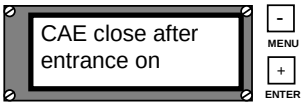


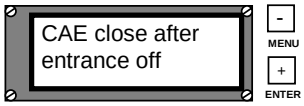


If selected adjustment in this point of menu is "toggel circuit = no" the impulses connected to A1/B1 resp. A2/B2 will be interpreted as clear open impulses. The adjustment "toggle circuit = yes" interprets the receiving impulses changing as open and close impulse. So for example will the first impulse received by the pull switch be valued as open impulse and the next coming impulse as closing impulse.

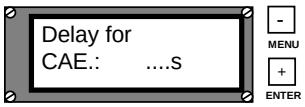
This function is possible with and without activated keep open time. At activated keep open time the door closes automatically after duration of preset time if not in the meantime already the second toggle current impulse has been activated.

8.3.3 Close after entrance function



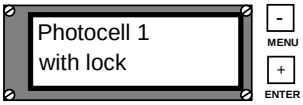


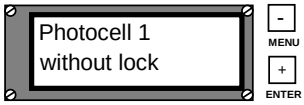
Upon request a close after entrance (CAE) function can be activated through the safety photocell 2. This means; If the photocell is interrupted after the door opened and is then free a closing impulse will be activated after an adjustable time of 2 to 15 seconds. A repeated interruption of the photocell leads to a new start of the delaying time of the CAE function. The safety photocell 2 with the standard door is mostly situated in the side columns. However, other setups are also possible. Of course may the CAE function also be activated by other impulse activators. But these must then be connected on the input clamps for the safety photocell 2.



If the CAE function has been selected the delay time of 2 to 15 seconds can be adjusted in this point of the menu.

8.3.4 Photocells



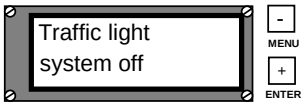


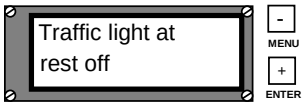
If a safety edge is installed always use the adjustment "SILI 1 with lock". If a prerunning photocell is installed use the adjustment „SILI 1 w/o lock“ since the door can be slowed down in a more controlled way because of the early detection.

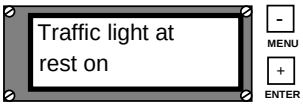
At the safety input 1 the prerunning photocell or the safety edge evaluation may be connected. This point of menu determines whether the door shall be locked or not after activating the safety feature. At the same time the slope of the brake ramp is adjusted.

8.3.5 Traffic light

(This point only shows if additional module 1 has been installed and activated, see 8.2.6)

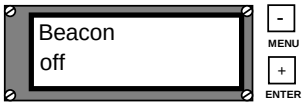


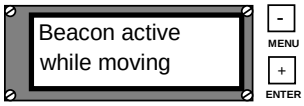


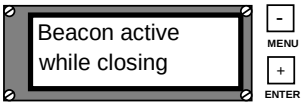


Various kinds of traffic control units make it necessary that with a closed door a traffic light will be completely switched off or otherwise remains on permanent operation. These two possibilities can be selected here. The point of menu „traffic light resting switched off“ means that as soon as the door is closed the traffic light is off. If the traffic light should remain in operation at closed door, select „traffic light resting switched on“ (see also advice in operation manual).

8.3.6 Flashing alarm light







A flashing alarm light may be warning as soon as the door gets into operation or only during the closing movement. Once the flashing alarm light is on during the up as well as during the down movement on the other hand only during the closing movement.



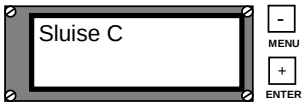
Traffic light as well as flashing alarm light may be used as prerunning control. This means; The flashing alarm light or the traffic light warn already before the start of the closing movement. The prerunning time is adjustable in this point of menu (0 to 15s). If no prerunning time is necessary adjust 0 as value. The prerunning time only starts after duration of the keep open time period. (Total keep open time = keep open time + prerunning time)

8.3.8 Airlock function







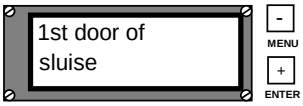


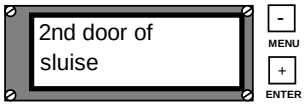
If two doors shall be used as airlock the desired airlock type may be selected here. The different airlock types roughly differ as follows:

Type A is planned for automatic impulse activators such as induction loops, radar detectors, etc.

Type B is planned for manual impulse activators such as push buttons, pull switches, etc.

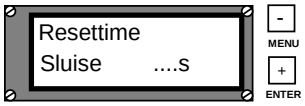
Type C recognizes via suitable impulse activators whether there is a vehicle within the airlock (transfer circuit condition). The kind of external impulse activators is at your choice.





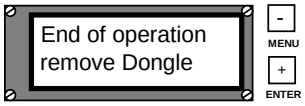
In the following point of the menu it is determined which door is the first door and which door is the second door of the airlock. This adjustment is only needed for the internal programming. For the run of the airlock function the selection is without any importance.

At an airlock there has always to be a first and a second door.



The reset time of the airlock is the time which after the run of the airlock function resets it to the original position (2 to 15s). Further information can be found in "Airlock control with FUS".

8.4. End of initial operation



Now only remove the programming plug from the plug X11 and the door can start working it's normal operation. Another activation of the menu button leads to a renewed start of the initial operation. Initial operation must always be finished with reaching this point of the menu.

8.4.1 Correction or optimization of the set values

If some of the values should be optimized or corrected after the first initial operation, the initial operation can be done over again. All correct set values can be jumped over with "MENU" button. If a point of the menu has been jumped over by mistake you may go through the whole menu with the "MENU" button. Data saving is done only by confirming with the „ENTER" button. A zero operation must always be done at each initial operation.

8.5 Errors during initial operation

If an error shows, firstly check the display in the control box for error an information. To correct the error you will find information in the error search manual at the appendix of the operation manual and in chapter 8.1.1 of this manual.

9 Operation

9.1 General information on frequency converter drive ACS 80

9.1.1 Introduction

The ACS 80 and the ACS80-2 drive has been developed for the application on industrial doors. It is TÜV certified and corresponds with CE regulations.

The drive contains of all necessary parts to drive the doors according all valid safety regulations. With up to 2 add-on modules many additional special functions can be realized. The characteristic of the frequency converter drive assures a soft start and brake function of the doors. This is an advantage at one hand avoiding hard blows for the mechanical part of the door and on the other hand allowing higher door speeds.

9.1.2 Safety advice

The safety features of the door are designed for the minimum protection level type 2 (DIN EN 12453):

„A limited group of persons will be trained for the operation of such door and the door is located in a public area."

Under consideration of such local situations, especially for pedestrian traffic during operation, additional safety devices resp. impulse activators may become necessary.

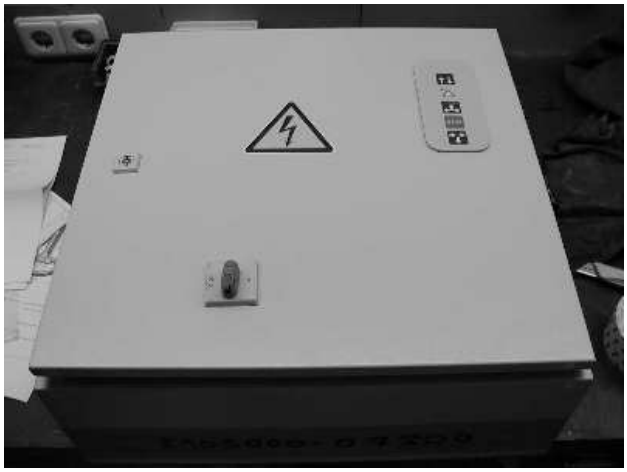
Please contact our sales engineers for further information and request advice at local site.

9.2 Basic design

9.2.1 Control panel with operation elements

The door drive is mounted on a plate in a metal control panel with the dimensions 500mm x 500mm x 210mm. The standard lock is a turnbuckle lock with a two-way key bit (other lock systems upon request). In the bottom are two connecting plates suitable for the installation of the power lines to the door and to the external impulse activators. For power connection a separate connection is provided. The inner door side of the control panel has a pouch containing the wiring diagram and operation manual, etc. The protection category of the control panel is IP54.

Built into the door of the control panel is the main switch and a key pad with 4 keys for the operation of the drive (see pict. 1). With the red main switch the drive is switched on and off and at the same time it serves as EMERGENCY-OFF switch. In the switched off position the main switch may be locked by means of a shackle type connector.



pict. 1: control panel exterior

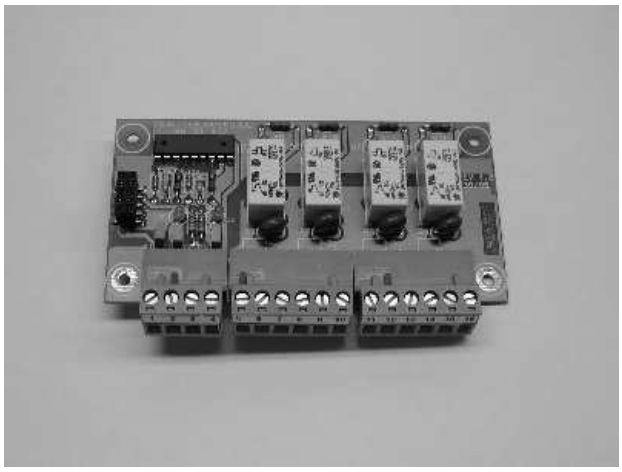


pict. 2: control box interior

The four keys „permanent open, open(Reset), stop and closed“ are provided for the operation of the door and to reset the drive system in case of error.

If the control panel is open the external main switch handle is without any function. The operation must be done via the internal main switch. Switching on the main switch is then also possible at opened door.

Inside the control panel (see pict. 2) at the right hand side is the electronic drive with the accompanying clear text display. The frequency converter incl. cooling device and filter device is situated at the left hand side of the drive. Up to 2 add-on modules may be optionally plugged onto the drive(see pict. 3). The contacting of the modules is done via installed plug-contacts at the downside of the module (X17 and X18 on the drive). Because of the identical design of both modules the arrangement is without any meaning. But each module has it’s own fixed function. Module 1 (left module, terminal block X17) is for traffic light and blinking light functions and module 2 (right module, terminal block X18) is for operation and lock information. For the function of these modules activation is necessary on the initial operation. (see initial operation manual).



pict. 3: Additional module

In the lower area of the control box a top hat rail is installed. If required this may be used for additional devices like induction loop detectors, radio receivers, etc. Left of the top hat rail is the main switch and 3 pre-fuses. The electrical connection of the three external conductors and if required the zero conductor is done directly at the main switch. The cable entry is lead from below through the cable nut in the middle of the control panel.

9.2.2 Drive terminal and frequency converter

The whole drive contains of a compact built printed board and a separately situated frequency converter. On the drive the following elements are accommodated.

- short circuit proof power pack for the micro processor drive,
- the micro processor with EPROM and operation elements with the clear text display,
- plug in terminals.

The frequency converter is installed at the left hand side of the control unit with an aluminum cooling body. The connection of the motor is done directly at the converter (terminals U2, V2, W2, PE). A power suppression filter is integrated in the cooling body. This filter serves to minimize repercussions to the power line. On top of the converter at the terminal X1 (with ACS 80) or RS 485 (with ACS 80-2) a connecting cable to the drive is installed. The green light on top of the converter is lit permanently for normal operation and blinking if malfunction occurs.

Because of the intermediate circuit capacitor there is still high voltage on the frequency converter up to 3 minutes after switching off the main switch.

9.2.3 Configuration of motherboard

The following plugs are intersections to the external components.

X1 connection 400 V power for drive

X2 brake

X3 fan for frequency converter (optional only at ACS 80 possible)

X4 connection for safety devices

X5 connection for add-on safety devices

X6 connection for external impulse activators

X7 connection for external impulse activators

X8 connection for external impulse activators

X10 connection for key pad or external display terminal

X11 intersection for air lock connection / dongle

X12 control line of converter / drive

X13 connection 24 V transformer for the drive unit

9.2.4 Function of keys

The control elements on the key pad contain the following functions:



Permanent open switch

If this switch is activated, the door opens up to it's maximum opening height/width and remains at this position until the permanent open function is activated again by the corresponding key being switched off. The permanent open position cannot be canceled by the button „Reset"- or „Close". The green diode shows the permanent open function and the display shows the message „key permanent open activated".

Message „door locked"



The yellow diode is lit if the door is locked. The reason for locking is shown at the clear text information. By activating the „Reset“ key the lock may be canceled again.



„Reset“ and „open“ key and error message

The red light shows error. The reason for error is shown at the clear text display. At the same time the key serves as „Reset“ key to cancel the lock again, and as „open“ key to give an opening impulse to the control unit.



„Stop“ key

With the „Stop“ key the door can be stopped at any position. At the same time the door gets locked and may only be set into operation again by activating the „Reset“ key. The stop function is shown as clear text message on the LCD display.



„Close“ key

With this key a closing impulse is given to the control unit. Since first of all the safeties are tested after each closing impulse, the closing operation starts after a short time delay only.

Beside the display at the control panel are 4 keys. With the key „Menu“ the keep open time during operation may be increased and with the key „Enter“ it may be lowered. (see 7.3.5).

9.3 Function

The track recording of the door is done via an incremental shaft encoder or via the limit switch. Therefore it is necessary for the door to operate a reference run in order to find the zero point after turning on the main switch or after activating the emergency lever (see 7.3.1). At doors with a limit switch the zero run must not be performed.

9.3.1 Turning on

The main switch is turned on. After a short time the yellow diode lights up „door locked“. On doors with incremental shaft encoder the display shows the message „Start of zero run = open“. The door is unlocked by activating the „Reset“ - resp. „open“ key. Afterwards the reference run is automatically operated. During the reference run the message „Reference run operating“ shows. Doors with a limit switch do not need a zero run.

Should the door be completely open during turn on position the message “close door by 20 cm“ shows and it must now be closed this far until the reference limit switch is left by activating the “close” button before operation of reference run. The door stops automatically after about 20 cm in order to avoid a manual overrun of the bottom end position by the operator. As soon as you get off the „close“ key, the message “Start zero run = open“ shows. By activating the „open“ button once the zero run starts. At the display shows the message "Reference run operation “.

After duration of keep open time or after a “close” impulse the door closes and is ready for operation again.

9.3.2 Open run of the door

An „open“ impulse is given by any impulse activator or the key board. The brake is relieved, the motor is driven by the frequency converter according a preset acceleration slope. The door opens with maximum speed up to the starting point of the brake ramp (adjustable in the initial operation menu or at the limit switch). Now the door is slowed down to about 1/6 of the opening speed by the frequency converter according the preset brake ramp/slope. In this slow speed (creep run) the door runs up to the preset upper end position. By reaching this point the motor turns off and the brake starts working.

Depending on the preset function, the keep open time starts now or the drive is waiting for a close impulse. The safeties are activated.

9.3.3 Close run of the door

After duration of the keep open time resp. at an close impulse all connected safety devices are tested first by the drive. If the safety test was successful, the safeties are activated, the brake is being lifted and the motor is driven by the frequency converter according a preset brake ramp/slope. The door closes with about 2/5 of the maximum speed up to the starting point of the brake ramp/slope (adjustable in the initial operations menu or at the limit switch).

Now the door is slowed down to about 1/6 of the opening speed according the preset brake ramp. In this slow speed (creep run) the door runs up to the bottom end position. By reaching this position the motor turns off and the brake starts working.



About 50 mm before reaching the bottom end position the safeties are turned off.

9.3.4 Clear text display

Malfunctions/errors and the activation of a safety are shown as clear text in the LCD display. During operation the preset keep open time and the number of operated door cycles is shown.

9.3.5 Keep open time and display of number of door movements

At the right hand side of the LCD display are 2 keys by which the keep open time may be increased in steps of seconds („Menu“ key) or decreased („Enter“ key). The adjustment level is from 0 to 99. The set value can be read in the upper line of the LCD display. In the lower line the number of door movements/cycles is shown. During the door movement the keep open time may not be changed.

9.3.6 traffic light function

If the menu point „traffic light resting on“ has been activated during the initial operation modus the relay for the traffic light control is permanently on (red traffic light phase). Activating the external impulse unit makes the function procedure for the traffic light active. This leads for the traffic light to change from the red to the green phase if door is completely open. During the door movement there is always the red phase active.



The internal terminal as well as the key pad at the control panel have no function on the drive for the traffic light phases since the direction can not be detected by these

keys. If the door is opened with the „open“ key at the key pad, the red phase remains active.

If the menu point “traffic light resting off” has been activated during initial operation modus the relay for the traffic light function is off. Only the activation of an external impulse unit leads to activation of the traffic light function. If the door is completely open the traffic light control changes from red to green phase. During the door movement there is always the red phase active. After the door closed completely an automatic turn off of the traffic light follows. Only a renewed activation of the external impulse unit leads again to activation of the traffic light function.



If the door is opened via the internal keys or the key pad at the control panel the traffic light is not activated since a direction cannot be detected. Only by closing of the door, after duration of the keep open time or park house time the traffic light will be activated to avoid crashes into the door by running traffic (red traffic light phase).



Activation of the permanent open function by pushing the permanent open key at the control box does not lead to an activation of the traffic light function! Only after deactivating the permanent open position the traffic light is activated by the closing movement of the door to avoid crashes with the door by the running traffic (red traffic light phase).

9.4 Safety features (see also chapter 9.7)

To avoid accidents the closing movement of an automatically running door must be carefully watched. For this purpose the following safety features are installed at the door.

Before every closing movement of the door the safety photocells are tested by the micro processor, turning them briefly off and on again. The processor evaluates the inputs and must come to the same impulse follow-up. Deviations from the set value mean the photocell is defect and the closing movement will not be started. The door may only be closed in hold to run operation. The check-up test is signaled by a briefly automatic lapse of the three LED´s (D5, D6 and D7) which show the safety inputs on the control. (see also 4.1)



During hold to run operation all safeties are out of operation. The door is running as long as the key at the control box is activated. At reaching the upper or bottom end position the door switches automatically off.

9.4.1 Safety edge (only with door type RR300, 390, 400 or optionally)

The safety edge serves for the safety of the closing edge. It is made of rubber profile which is assembled on the main closing edge of the door. Deformation of the rubber profile activates an electrical impulse. This impulse stops the door immediately. Afterwards it runs independent of the preset opening height/ width up to the reference limit switch and is locked there. Locking is shown by the yellow lit diode at the key pad. Activating the „Reset“ key cancels the lock position. Now the door may pick up it’s normal operation again.

To avoid malfunction or fail function of this safety device the control unit checks it before every closing movement. An interruption of the safety edge as well as broken cable, short circuit or to heavy deviation of the value of resistance from the nominal value are detected by the control unit and signaled by a red light at the control panel. Now the door can be closed again only via the „CLOSE“ key at the control panel in „hold to run mode“.

9.4.2 Stationary safety photocell

The stationary safety photocell is installed in the lower area of the door. Also here the correct function is safeguarded by the control unit. If the photocell is activated in open door position a reset and renewed start of the keep open time follows. Activation during the closing movement leads to an immediate stop of the door and following reopening up to the reference limit switch (reversing). In this end position the door is not locked like with safety edge but closes automatically after duration of the keep open time.

9.4.3 Prerunning photocell (only at door types RR 350, 355, 450 + RT 800)

The prerunning photocell runs about 100 mm below the bottom profile. This photocell operates during all the open and closing movements of the door. In the bottom end position of the door it immerses in the bottom profile. Activating the prerunning photocell has the same effect as activating the safety photocell if in the initial operation menu it has been set as regular safety photocell. Since a compulsory activation of the prerunning photocell is made by the bottom profile shortly before reaching the bottom end position of the door it is, of course, in off position within this area. This photocell is also safeguarded for it's correct functionality before every closing movement of the door. Switch-off of the photocell is made as soon as it hits the bottom position.

9.4.4 „Stop" button

By activating the „Stop" button the door can be stopped in every position. A following „Reset" - impulse brings the door into regular function again.

9.4.5 Running time safeguard

To ensure a safe automatic operation of the door the running time of the opening and closing movements are safeguarded. If the corresponding end position is not reached within the preset running time after an opening or closing impulse, the control cuts the motor off and the red error lamp lights up. At the same time a clear text message on the display „running time exceeded" shows. Only after confirming with the „Reset" button a renewed impulse brings the door in operation again.

9.5 Power failure

Upon power failure the door remains in it's current position. After return of power the door can only be operated again by activating the „Reset" button and making a reference run (see 7.3).

9.6 Emergency manual operation

Upon power failure, repair work and maintenance work it may become necessary to open the door manually.

Before activating the emergency lever/crank always switch off the main switch. After activation of emergency lever the door can partly open or close by itself !

The brake of the drive motor may be released via a bowden cable by the emergency lever at the side column. Depending on the design of the balance weight the door leaf can partly open or close by itself. Now you may reach with your hands below the bottom rubber profile in the middle of the door pushing up the door leaf. Doing this the door curtain rolls up onto the top roll. With the RT 800 the segments are pushed up accordingly. To reach regular function of the door again the emergency lever must be put back into it's original position. After that the control unit may be switched on again. Activating the „Reset" button leads to the start of zero action run. After finishing the zero action run the door is ready for normal operation again. With the RT 150 there is only an emergency crank which must be plugged in at the front side of the motor. By winding the door may be opened or closed.

Should the emergency lever still be active by turning on the main switch, the red error lamp lights up at the key pad and at the display appears the clear text message „emergency lever active". This message can only be confirmed after bringing the emergency lever into it's original position or by removing the emergency crank. The confirmation is made by activating the "Reset" key at the control panel. If activated again the reference run is initiated (see 7.3.1). After that the door is ready for operation again.

The door is not designed for permanent manual operation.

9.7 Safety features

9.7.1 Temperature contact

For the safety of the drive motor 3 temperature contacts are incorporated in it's coil. If these contacts react the motor gets to hot and is immediately cut off by the drive. At the same time the red lamp at the control panel starts to light up. Now it is necessary to switch off the drive and let the motor cool down. The door as well as the drive should then be checked for mechanical or electrical faults. Only after the motor cooled down the drive should be started again. If there is no error the drive will automatically be reset and regular operation may start again.

9.7.2 Fuses

Three mains fuses 10A are installed in the control panel.

As spare only fuses of the same type are allowed.

9.7.3 Crash sensor

With some door types it is possible to install a knock out capability at the bottom profile to keep damages on crashes at a minimum. If the door has been crashed into, the crash sensor is activated and the door stops immediately. The error lamp at the control panel lights up. At the same time the message „door crash, please close" shows up at the display. Now the dislocated bottom profile must be repositioned again. There are two different versions to do it.

Door type RR 355, 450 :

The door must be completely closed in hold to run function. The close key at the control panel must be as often activated until the knock out facility is repositioned again. A renewed „OPEN“ impulse puts the door in operation again.

Door type RR 300 and 390:

The knock out facility must be manually repositioned into the side columns again. Thereafter the error message can be confirmed and the door is ready for operation again.

After every crash procedure the door must be checked thoroughly for possible damages.

9.8 Mechanical lock (only with RapidTech 800)

The retaining device holding the bottom panel is designed in such way that pushing up the door leaf in closed position is only possible by about 150mm.

Further exists an additional manual lock at the drive side in the lower side column. With this lock the door may be securely locked up (pict. 4).

Through activation of lock the door is stopped and a yellow lock signal lights up at the key pad. On the display the clear text message „External key Stop activated“ shows. This message can only be confirmed after unlocking with the „Reset“ key. After that the door is in it’s regular operation modus again.



pict. 4: mechanical lock

(optionally with RT 150)

To surely avoid the push-up of the door leaf an optional lock to be manually operated is available. If the push bolt is inserted the door can not be started. A yellow lamp lights up at the key board and at the display is the message “external key stop activated”. After the door has been unlocked, the message can be confirmed with the “reset” key. The door is then ready for normal operation again.

9.9 Slack rope switch (only with RT 150)

After the slack rope switch has been activated the door must as soon as possible be checked by authorized personnel! The pull string goes off center at the pulley through the missing (briefly) power. The door must not be set in operation again!

To safeguard the pull devices the RT 150 is furnished with 2 slack rope switches. One slack rope switch is operated automatically if there is no power on the strap in the set direction. Upon activation the message “ external key stop activated” shows at the display. After error repair this message may be confirmed with the key “reset”. Now the door is ready for operation again.